Impact! CNC Tornado – CNC Router

Machine specification & asset record · OEM build: Excitech E2-SYNTEC (2017)

Saturn Electrical
Warwick QLD

Machine Overview

Make / Model	Impact! CNC **Tornado** flatbed router (Excitech-built)
Year of build	~2017 (drawings 20-03-2017, transformer 08-2017)
Configuration	Moving gantry, 3-axis, dual-drive Y (Y1/Y2)
Drive system	Helical rack & pinion (X/Y), linear guide rails all axes
X travel	~1,554 mm (−12 to 1,542 per panel notes)
Y / Z travel	To confirm
Workholding	Vacuum hold-down table, 4 zones, individual valves + gauge
Accessories	Coolant mist atomiser, dust hood w/ brush skirt, tower light, MPG handwheel, 2× E-stop
Serial No.	
Asset No. / Location	



Spindle

Model	GDZ120×103-5.5, async. 3-phase
Power	**5.5 kW**
Voltage / Current	380 V ±10% / 9.5 A
Frequency / Speed	300 Hz / **18,000 rpm max**
Cooling / IP	Air-cooled / IP50
Collet	ER32 (likely) – confirm

Spindle VFD (UF1)

Model	Delta **VFD055E43A** (VFD-E)
Input	3PH 380–480 V 50/60 Hz, 14 A
Output	3PH 0–480 V, 13 A, 9.9 kVA
Rating	5.5 kW / 7.5 HP
Freq. range	0.1–599 Hz
Supply path	FU1 fuses → KM1 contactor → UF1

CNC Control

Controller	**Syntec 6MD** w/ MPG pendant
I/O boards	Syntec TB16IN-5.x (16 in), TB16OUT-5.1 (16 relay out)
DC supplies	2× Delta PMT 24 V DC
Comms	LAN, RS485

Servo Drives – Yaskawa Σ-7 (200 V)

Axis	Drive	Size
X	SGD7S-5R5A10A002	750 W
Y1	SGD7S-5R5A10A002	750 W
Y2	SGD7S-5R5A10A002	750 W
Z	SGD7S-2R8A10A002	400 W
Unlabelled	SGD7S-2R8A10A002	400 W ?

Power Supply & Switchgear

Supply	**415 V 3PH 50 Hz + PE**, main isolator QS1
Servo transformer	Lapuele LSG-4KVA, 415→220 V Y/Y, 5.5 A / 10.4 A
Vacuum pump (M9)	QF1 GV2-ME16C (9–14 A) + KM3
Dust extract. (M10)	QF2 GV2-ME16C (9–14 A) + KM4
Fusing	MRO RT18-32, 10×38, 32 A
Other	Schneider TeSys contactors, mains EMI filter

Capability / Materials

Timber, MDF, ply, plastics, ACM	✔ Primary use
Aluminium (≤6 mm sheet)	✔ Light passes, mist coolant, mech. clamping
Brass / copper	◖ Light duty only
Mild / stainless steel	✘ Not suitable – use plasma/laser

Notes:

- Servo drives are **MECHATROLINK-II** type ("10" interface) – not step/dir. Any controller retrofit (e.g. Masso) needs drive replacement (SGD7S-xxxA00A) or a full servo package. Confirm from drive side labels.
- Do not run aluminium swarf through the wood dust extractor – fire risk. Isolate extraction when cutting metal.
- Maintenance flags (Nov 2026): surface rust on spindle nose, collet nut & racks; cabinet fans/filters dusty. Clean, oil racks, service fans.
- Job origins (G54 / G58) and Z references currently on whiteboard – transfer to a written setup sheet.